

Architecture Modeling of Wind Farm Architecture- Security Zones and Trust Boundaries

Zone	Representative components	Trust boundary / enforcement
OEM domain	OEM SCADA server; PCN	OEM firewall mediating in-bound/outbound connectivity
Vendor domain	Vendor control room	Vendor firewall separating third-party administrative boundary
Customer domain	Customer control room; customer grid control server	Customer firewall enforcing customer-side perimeter
Wind-farm control room	Wind-farm control room server	Wind-farm control room firewall separating control room from other domains
Turbine local control	Master HMI; Slave HMI; PLC; Distributed I/O; sensors/actuators	Local control network boundary (field domain)
External transit	Internet; customer WAN; VPN links	Transit networks enabling cross-zone connectivity

Architecture Modeling of Wind Farm Architecture- Cross zone conduits-Communication Paths

Conduit		Endpoints (zones/as-sets)	Security relevance for modeling
Vendor	VPN	Vendor domain → OEM domain / OEM network center	Enables remote vendor sessions; VPN termination and firewall policy mediate reachability
Customer VPN		Customer domain → customer-side endpoints	Defines protected remote access channel for customer operations
Customer WAN		Customer domain ↔ wind-farm control room	Cross-domain operational visibility/control; boundary policies constrain access
Control-room → field control path		Wind-farm control room → turbine local control (HMI/PLC/I/O)	Mediates operational control and configuration actions reaching the physical process

Architecture and Authorization- Formal Definition

Conduit		Endpoints (zones/assets)	Security relevance for modeling
Vendor	VPN over Internet	Vendor domain → OEM domain / OEM network center	Enables remote vendor sessions; VPN termination and firewall policy mediate reachability
Customer	VPN	Customer domain → customer-side endpoints	Defines protected remote access channel for customer operations
Customer	WAN	Customer domain ↔ wind-farm control room	Cross-domain operational visibility/control; boundary policies constrain access
Control-room	→ field control path	Wind-farm control room → turbine local control (HMI-PLC/I/O)	Mediates operational control and configuration actions reaching the physical process